

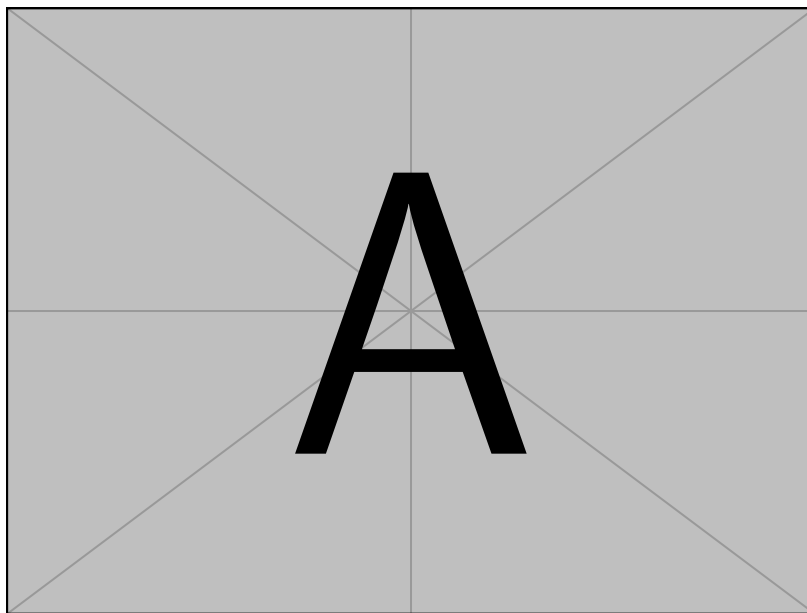
DATE: 00.00

FIGURE 00.00X.1: Example image for templating.

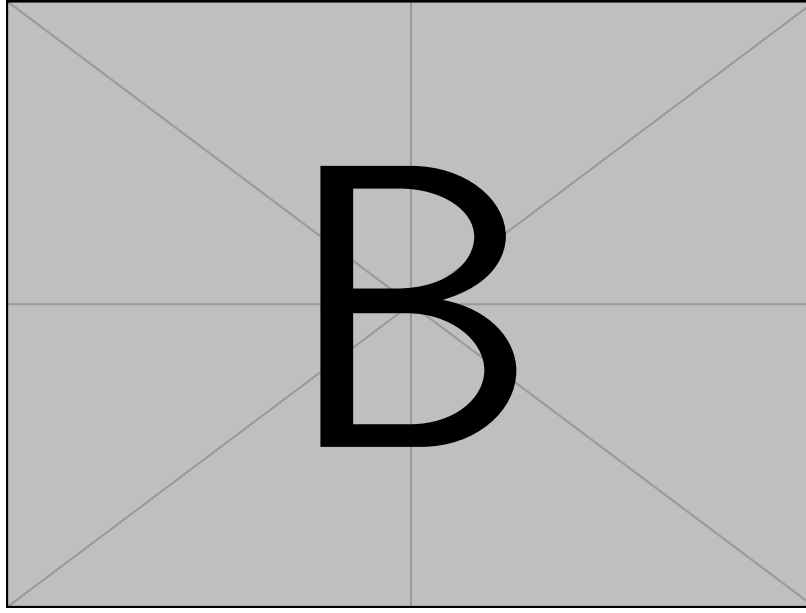


FIGURE 00.00X.2: Long caption. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum..

DATE: 03.11

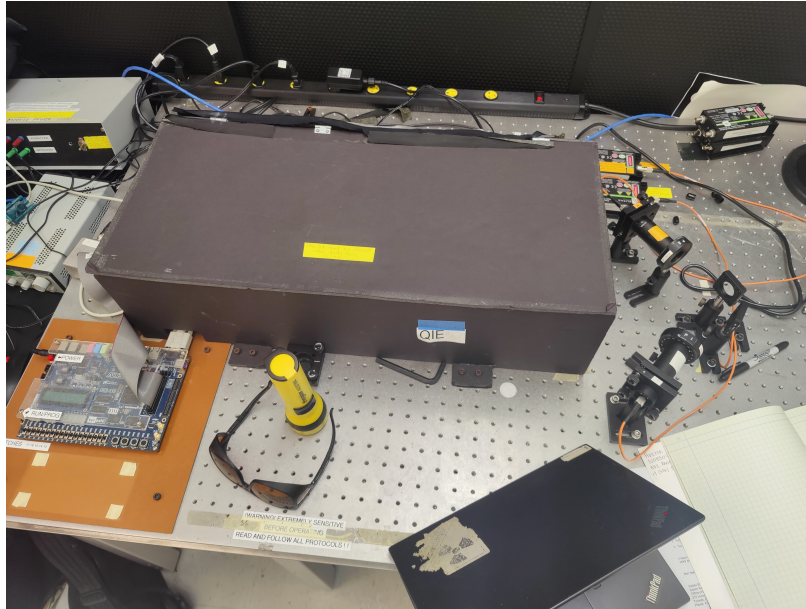


FIGURE 03.11A.1: Initial experimental setup prior to any changes. All components seem to be in place. Noticed no caps on extra APD detectors.



FIGURE 03.11A.2: Experimental setup within black box. Laser and powersupply and all other relevant components outlined in P03.11A turned off prior to opening.

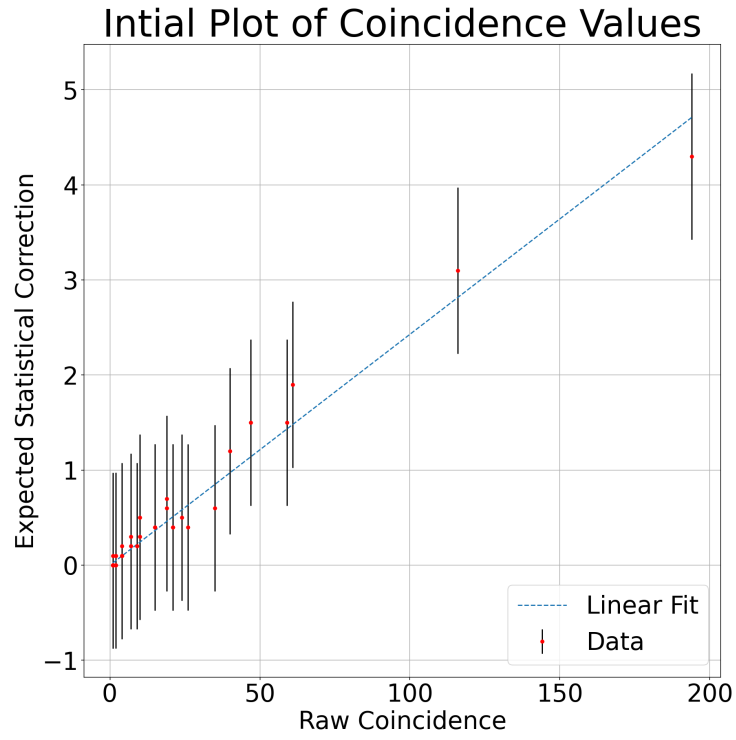


FIGURE 03.11A.3: Initial fit of coincidence values and expected statistical correction. Slope found is given by $\tau = 0.0243 \pm 0.0006$ with a $\chi^2_{\text{red}} = 0.03$.

DATE: 03.14

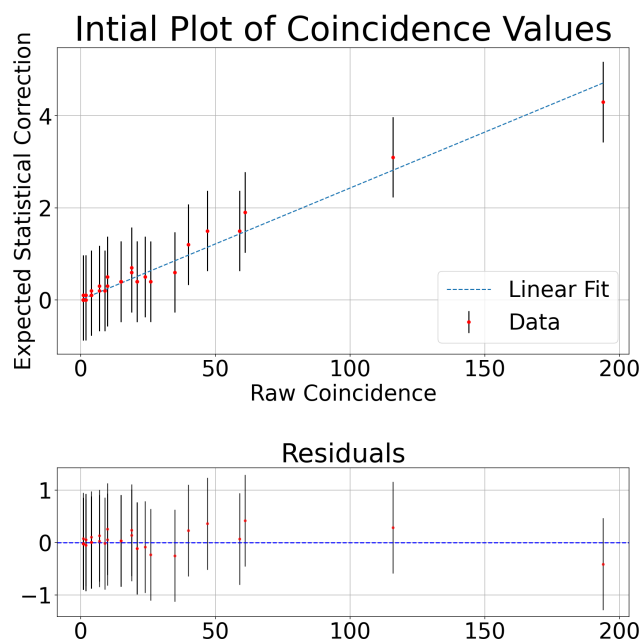


FIGURE 03.14A.1: Same fit of data as in Fig. 03.11A.3 but with residuals.

DATE: 03.18

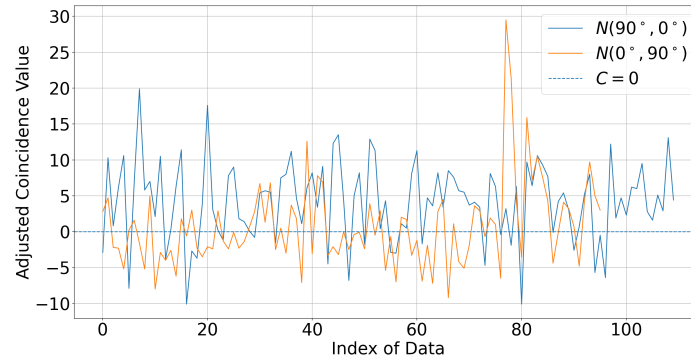


FIGURE 03.18A.1: Plot of series of $N(90^\circ, 0^\circ)$ and $N(0^\circ, 90^\circ)$ data points to illustrate variation in data.

DATE: 03.19

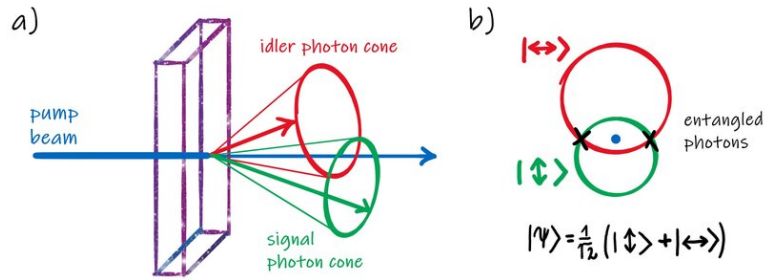


FIGURE 03.19D.2: Diagram of Spontaneous Parametric Down Conversion (SPDC) process. **Source:** Applications of single-photon technology - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/The-SPDC-cones-a-Type-II-process-b-Polarization-entangled-photon-pair_fig10_360782305 [accessed 19 Mar 2025].

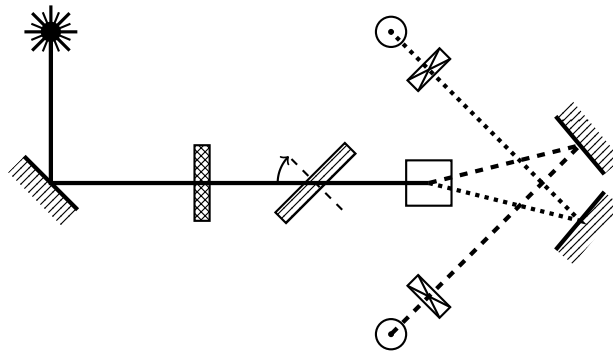


FIGURE 03.19E.3: Experimental setup (405 nm).

DATE: 03.26



FIGURE 03.26A.1: Half Waveplate and Quartz Plate apparatus.



FIGURE 03.26B.1: 405 nm Diode Laser (beginning of setup.)



FIGURE 03.26C.1: Avalanche Photodiode (APD) detector and Polarizer Setup.

DATE: 03.22

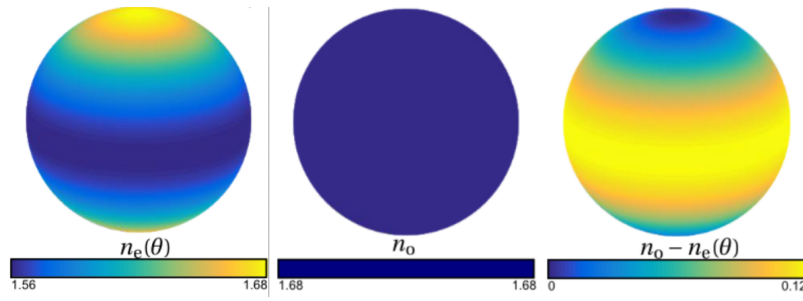
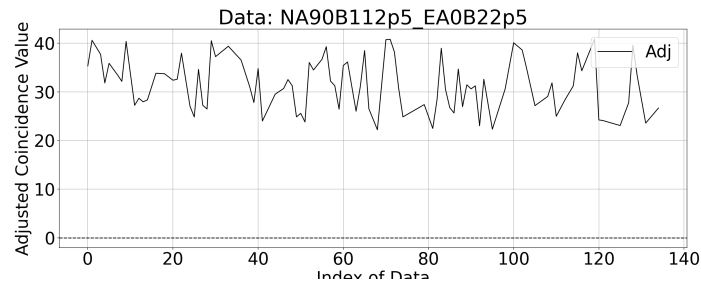
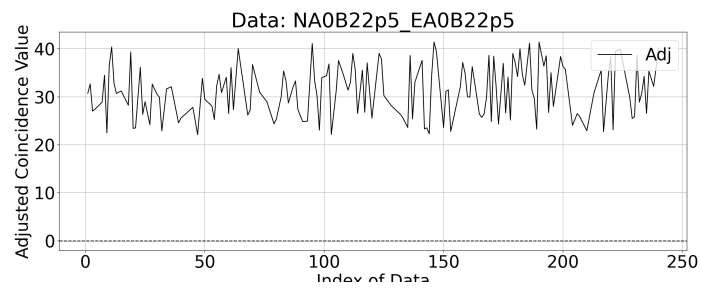
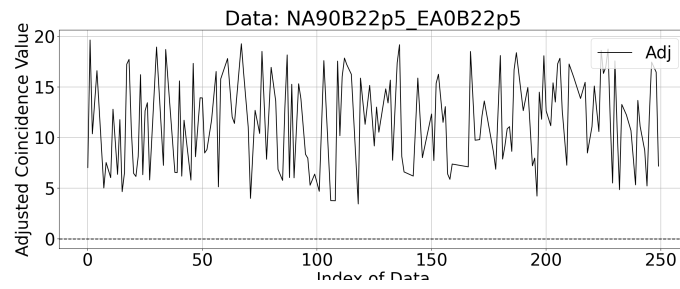
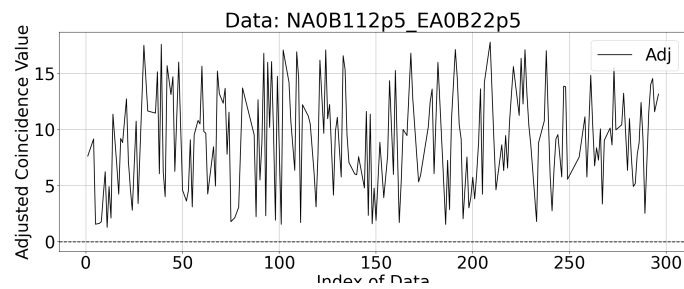


FIGURE **03.22A.1**: The extraordinary and ordinary refractive indices (and their difference) as a function of direction for beta barium borate (BBO) at $\lambda = 500$ nm ($n_o = 1.68$ and $n_e = 1.56$) [5].

DATE: 03.27**FIGURE 03.27A.1****FIGURE 03.27A.2****FIGURE 03.27A.3****FIGURE 03.27A.4**

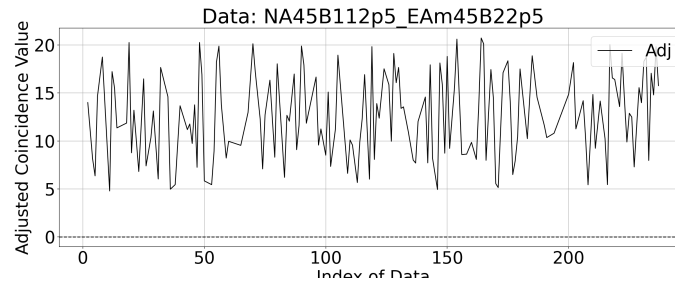


FIGURE 03.27B.1

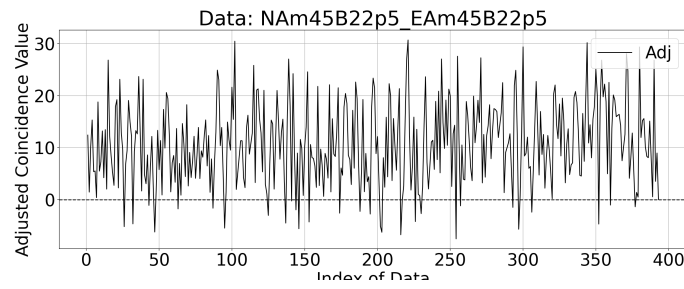


FIGURE 03.27B.2

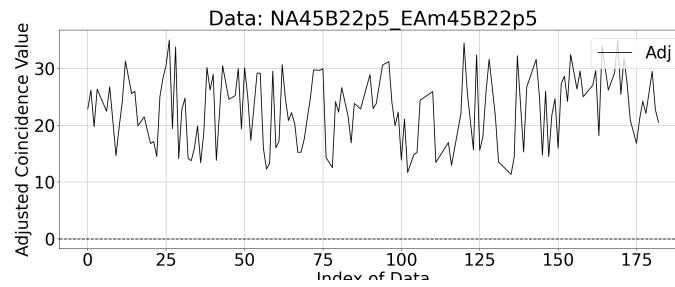


FIGURE 03.27B.3

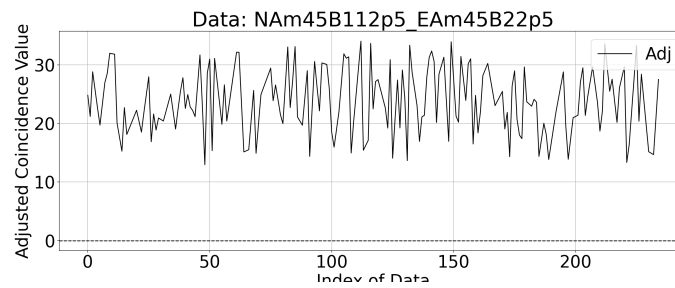


FIGURE 03.27B.4

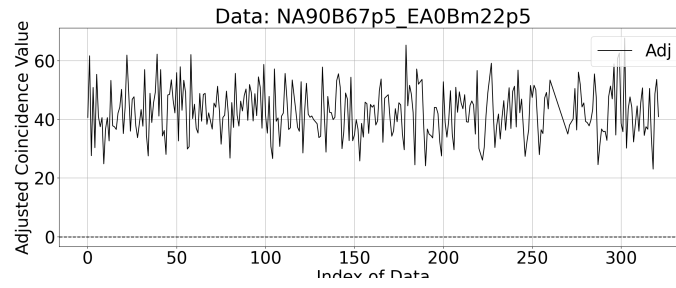


FIGURE 03.27C.1

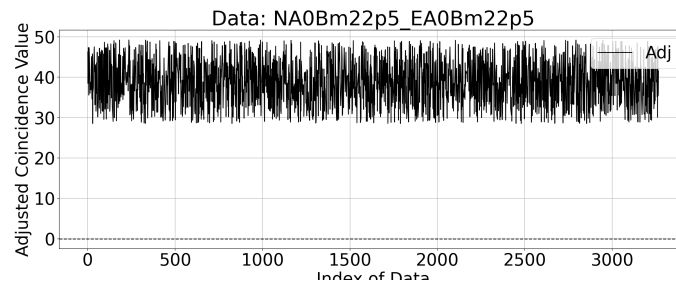


FIGURE 03.27C.2

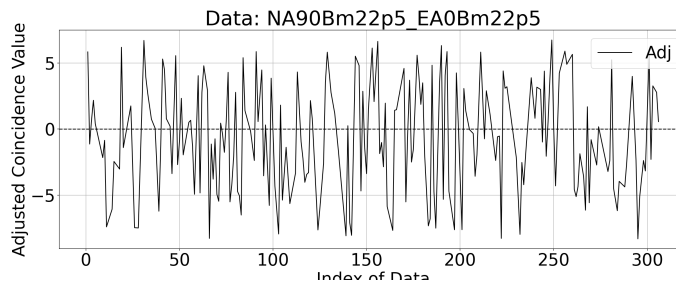


FIGURE 03.27C.3

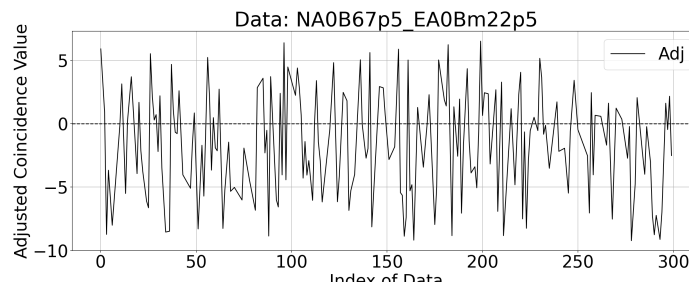


FIGURE 03.27C.4

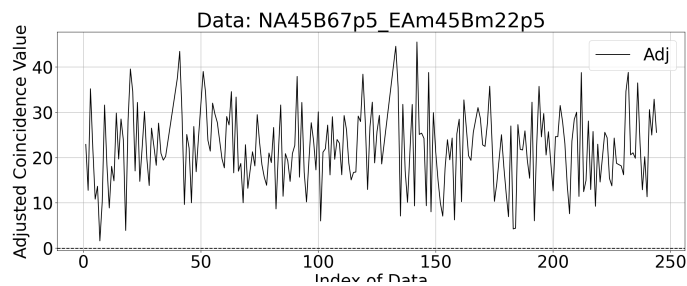


FIGURE 03.27E.1

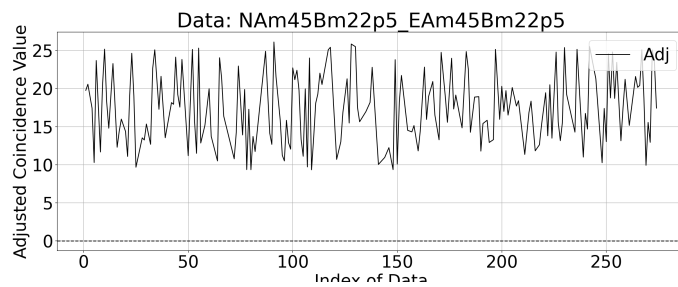


FIGURE 03.27E.2

DATE: 03.25

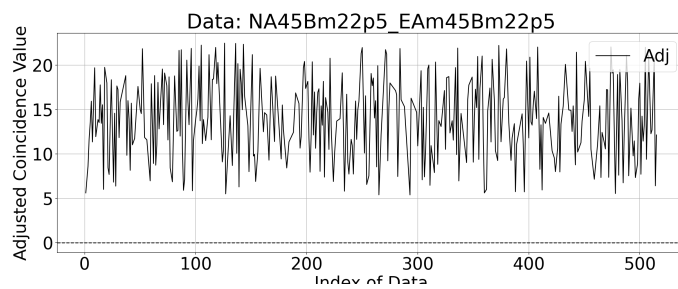


FIGURE 03.25A.1

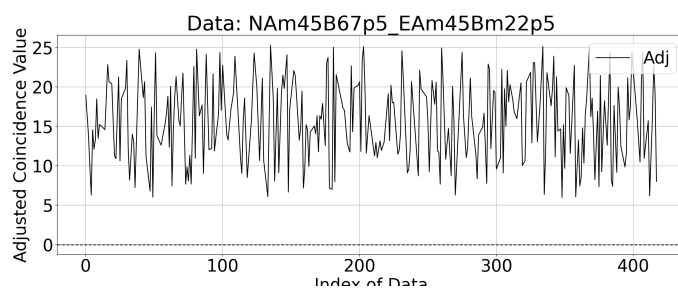


FIGURE 03.25A.2

DATE: 03.30

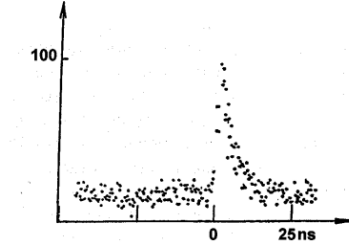


FIGURE 03.30B.1: Illustration between delay time and coincidence rate [7] known as the *Time Delay Spectrum*. The flat background corresponds with accidental coincidences between uncorrelated photons emitted by different atoms and scales with the square, N^2 , of the cascade rate. The peak, whose area scales with N , corresponds with the correlated photons and gives the coincidences to be measured.

DATE: 04.03

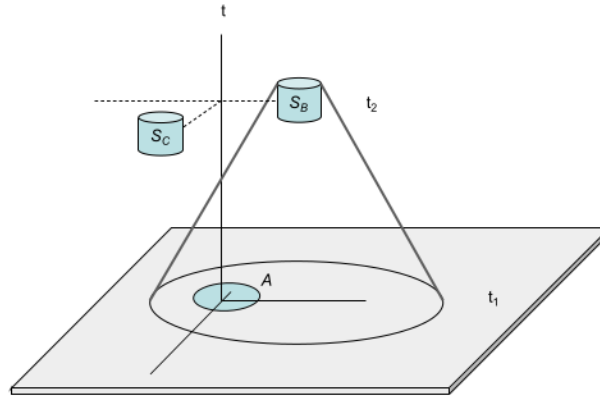


FIGURE 04.03A.1: Space like separated regions as S_A is not in the backward lightcone of S_B . Figure from Prof. Sipe's PHY491 notes [12].

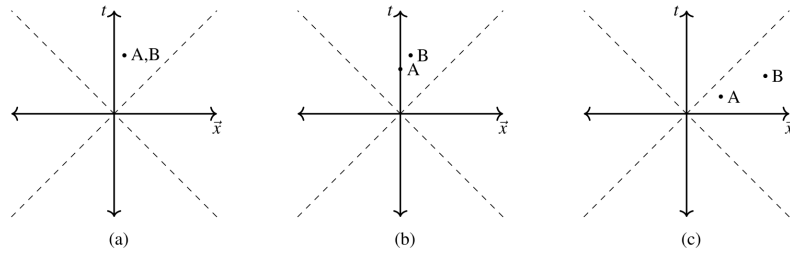


FIGURE 04.03B.1: (a) Two events/interactions at the same space-time location are local. (b) Two time-like events separated by a distance preserve Einstein's locality condition. (c) Two space-like separated events are nonlocal. Figure and caption from [3].

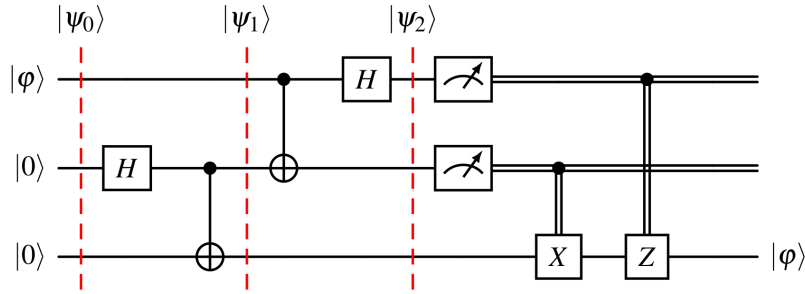


FIGURE 04.03C.1: Quantum teleportation circuit as described (with modification) by Nielson and Chuang [2]. The first Hadamard ($\hat{H}$) and CNOT gates describe the creation of a Bell state. Figure and caption from [3].

DATE: 04.04



FIGURE 04.04A.1: APD and polarizer setup. This is the same figure as Fig 03.2C.1. The main thing to note is that the fibers are not adequately spaced before the measurement devices.

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